

Question:

A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient’s X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient’s condition?

Answer Choices:

- A. It activates TLR5 on the surface of macrophages.
- B. It primarily induces the Th1-cell response.
- C. Peptidoglycan is its major antigen that induces an immune response.
- D. This pathogen evades the immune response by encapsulation.

Reasoning + Answer:

<think> The patient is diagnosed with community-acquired pneumonia , which is often caused by bacteria such as Streptococcus pneumoniae . The immune response elicited by this pathogen is a result of the innate and adaptive immune response . The correct answer is related to the characteristic of the pathogen that causes community-acquired pneumonia . Peptidoglycan is a major component of the cell wall of Gram-positive bacteria , including Streptococcus pneumoniae . The immune response to peptidoglycan involves the activation of the innate immune system , including the recognition by toll-like receptor 2 (TLR 2), not TLR 5 , which is specific for flagellin . The adaptive immune response to community-acquired pneumonia is primarily through the Th 17 -cell response , not Th 1 -cell response , which is more associated with infections such as Mycobacterium tuberculosis . The pathogen that causes community-acquired pneumonia does not evade the immune response by encapsulation , which is a characteristic of encapsulated bacteria such as Streptococcus pneumoniae , but rather the capsule helps the bacteria to avoid being phagocytosed . </think> <answer> C . Peptidoglycan is its major antigen that induces an immune response . </answer>